VF Asia Footwear Laboratory Testing Manual

Test Method for Whole Shoe Flex - Water Proof Test	Test method # : VF- G04 REF #: TM-R-054 Page 1 of 9
Date Issued : May 20, 2016 Revision # : 3 Reason for revision: Modify the word on marking -" if using an insole as reference is NOT practical, use the outsole".	Created by : Nila Babia Erazo

PURPOSE AND SCOPE: To determine the ability of a shoe to withstand repeated flexing or bending to a 30 degree angle while submerged in the water.

REFERENCES: SATRA TM 77-1992

EQUIPMENT AND MATERIAL:

1. Shoe Flexing Waterproof Tester
2. Whole Shoe template
3. Sponge
4. Template
5. Masking tape ((½" wide)
6. PE plastic
7. Wire detector
8. Wire mesh
9. Water at normal temperature
10. Allen wrench and wrench
11. Footwear

SAMPLE PREPARATION: Option 1

1. Condition the test sample in a standard controlled environment at 23 +/- 2 °C and at 55 +/- 5 % RH for at least 48 hours to allow the cement to crosslink completely.

2. Remove all hang tags, laces, inserts, and tissue from dry footwear.
3. Draw a line down the length of the insole of the shoe from the center of the heel towards the center of the toecap. ***Note: If using an insole as reference is NOT practical, use the outsole.***

4. Measure along the line the approximate distance given in table 1, from the heel end of the line and mark a point to the nearest millimeter

5. Mark the second point on the line drawn which is 5 ± 1 mm closer to the heel.

6. Draw a line across the width of the insole board or **outsole**, which passes through the second point and at 90 ° to the line drawn. This line is regarded as the flexing line of the shoe. Refer to figure 1.

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6. Use a template suitable to the size of the specimen (the front part of the template should be round so that it can be pressed against the toecap).
7. Place the template on the wire mesh and then push against the toecap.
8. The edge of the back part of the template shall coincide with the flexing line drawn on the insole. Tie the shoe and then put it in the testing area.

REPETITIONS: Test half or one pair of shoes per type being evaluated.

Table 1

Children's		Men's **(Note: for Women's should use 1 size lower)**	
Size	Heel/ Ball length on insole , mm	Size	Heel/ Ball length on insole , mm
4	85	1	143
5	91	2	149
6	97	3	155
7	102	4	160
8	108	5	166
9	114	6	172
10	120	7	178
11	126	8	184
12	131	9	189
13	137	10	195
		11	201
		12	207

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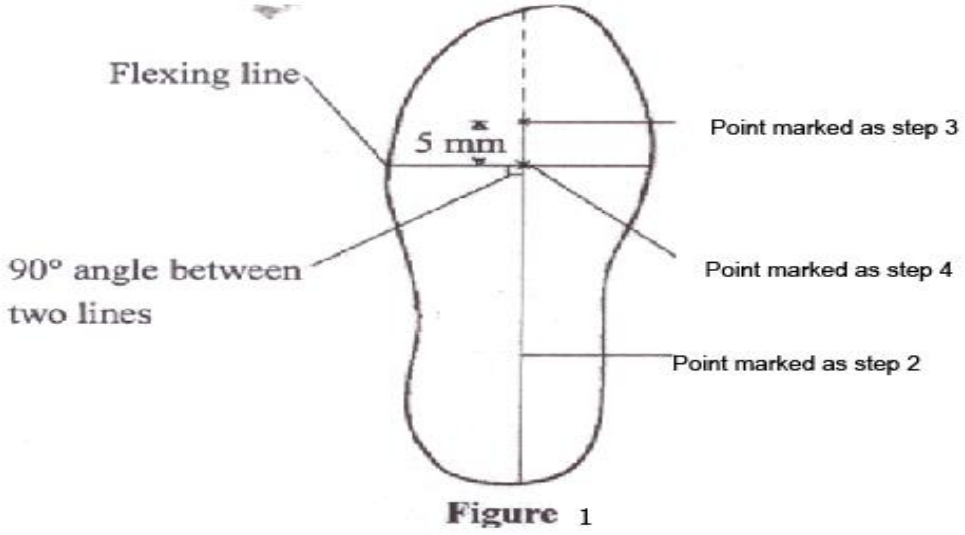


Figure 1

TEST PROCEDURES: Testing shall be conducted in a standard laboratory environment of 23 ± 2°C and 55 ± 10 % RH.

1. Set the flexing angle at 30° .
2. Set the flexes per minute between 70 ±10
3. Open the left door of the machine.
4. Loosen the nut and the fixed block 1 and then adjust the nut to an angle of 30 °
5. Use the wrench to rotate the link to tighten the nut. Lock then fixed the block 1.
6. Loosen the nut on the eccentric wheel
7. Adjust the bolt until the slider coincides with the angle 30 ° scale.
8. Tighten the nut through clockwise direction.
9. Put the shoe in the machine.
10. Set the distance between the toe cap clamp and the heel clamp in accordance with the size of the shoe selected.

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11 Fix the specimen to the heel seat and then set the hook to make it hold the specimen vertically

12 Adjust the handle wheel, nut and top plate to a suitable plate so that the toecap is on the non-sliding plate and the flexing line coincides with the edge of the non-sliding plate. Rotate the handle wheel until the platen presses against the template inside the toecap.

13 Insert two ends of the conducting pins into the corresponding jacks, and then connect two other ends to the wire mesh inside the toecap.

14 For the sensitivity of water leakage impedance, switch on the power button. Then switch on the calibration button to the calibration position. Insert the red and the black probes of the universal meter into the corresponding meter into the corresponding Ω jacks. After which set the penetration adjustment to the required Ω step.

15 Fill the container with water at normal temperature and adjust the level as specified per style. Should use a colored water such as red, blue, or black as the test water in order to help identify the point of water penetration.

16 Set the speed adjuster to the required speed step.

17 Switch the test/ calibration button to the test position.

18 Set the "shut-off" counter to the appropriate number of cycles then set the cycle counter to Zero.

19 Press the "Start Button "to initiate the testing but be sure that the BUZZER button is pressed before pressing the START BUTTON.

20 When the required number of cycles has been reached, the machine stops automatically. (Note: If the machine has no provision of automatic stop, press the STOP button, and then press the POWER button off). Remove the specimen.

21 Asses the result by checking inside the tested shoe sample.

- Inspect any presence of water. Wipe with white tissue paper on the edges to double check any penetration of water.
- If leak(s) are detected, note area(s) where water penetrated or wicked. Mark wet area(s) with pen – use silver ink for black or dark colored material.
- To ensure that the area of water leak is properly identified, cut with knife along the bonding position of the shoe and observe the position and extent of water penetration.

22 Record the number of cycles at which the water penetrated.

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REPORTS: The number of cycles at which water penetrated.

SAMPLE PREPARATION / Option 2: TM-R-054

1. Condition the test sample in a standard controlled environment at 23 +/- 2 °C and at 55 +/- 5 % RH for at least 48 hours to allow the cement to crosslink completely.

2. Remove all hang tags, laces, inserts, and tissue from dry footwear.

3. For adult footwear, draw a water level line with a silver ink pen on the footwear so the water line will be set at 60% of the height of the lowest point on the top line as measured from the bottom of the outsole (Figure 1). For kids' footwear, draw a water level line with a silver ink pen on the footwear so the water line will be set at 50% of the height of the lowest point on the top line as measured from the bottom of the outsole (Figure 2). For both adults and kids footwear, if the lowest point on the top line is greater than 195 mm, the water height will be set at 120 mm (Figure 3).

   Note: The shoe should be constructed such that it is waterproof to a point 19mm above the water level line. This point is known as the "High Water Mark" of the shoe**.**

TEST PROCEDURE: Testing shall be conducted in a standard laboratory environment of 23 ± 2°C and 55 ± 10 % RH.

1. Set the flexing angle at 30° . Follow steps 3-8 of Option I to adjust the angle.

2. Set the flexes per minute between 70 ±10.

3. Reset automatic counter to zero.

4. Insert footwear into reservoir and secure at flex point or ball area using rubber footpads. Clamp at the heel *(Note: Clamped loosely so that the shoe is not pulled against the toe spring when the flex angle of the machine goes to zero. Ensure that the machine equipped with a long clamping arm, a sliding heel clamp to allow proper clamping). Figure 4.*

5. Secure the tongue and other upper components to prevent premature failure *(Note: tape or plastic wrap can be used to prevent water from splashing onto the materials , such as gore stretch fabric , located above the water proof line. Tape or plastic wrap can also be used to seal the top collar of the boot in order prevent the splashing of water into the shoe).*

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6. Fill reservoir with clean water such that at the maximum bend/flex of the shoe, the water level will be at the water line or completely covering the flex area of the vamp, whichever point is lower- refer to figure 5. This will ensure that water will not enter the footwear by splashing over the topline.

7. Set the speed adjuster to the required speed step.

8. Switch the test/ calibration button to the test position.

9. Set the "shut-off" counter to the appropriate number of cycles then set the cycle counter to Zero.

10. Press the "Start Button "to initiate the testing but be sure that the BUZZER button is pressed before pressing the START BUTTON.

11. When the required number of cycles has been reached, the machine stops automatically. (Note: If the machine has no provision of automatic stop, press the STOP button, and then press the POWER button off). Remove the specimen.

12. Asses the result by checking inside the tested shoe sample :
    - Inspect any presence of water. Wipe with white tissue paper on the edges to double check any penetration of water.
    - If leak(s) are detected, note area(s) where water penetrated or wicked. Mark wet area(s) with pen – use silver ink pen for black / dark colored material.
    - To ensure that the area of water leak is properly identified, cut with knife along the bonding position of the shoe and observe the position and extent of water penetration.

13. Record the number of cycles at which the water penetrated.

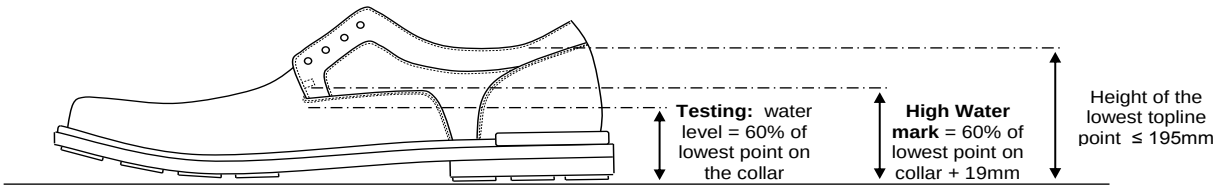
REPORTS: The number of cycles at which water penetrated.

Notes: For Options 1 *and* 2:

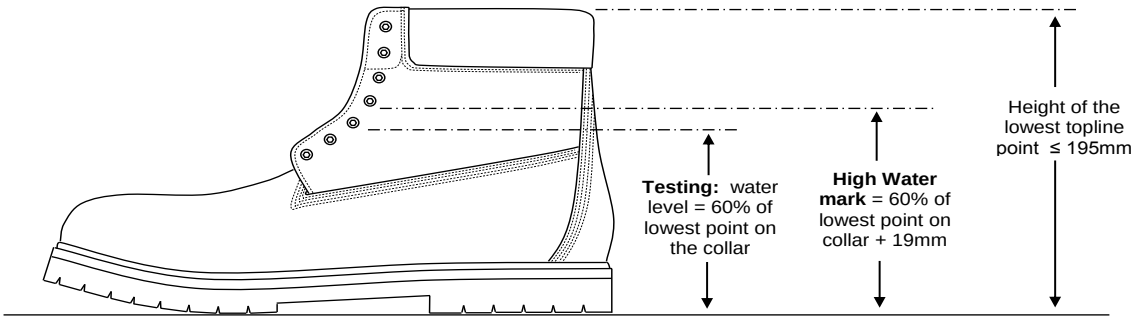
1. If the machine has no provision of conducting pins, manually monitor every 5,000 flexes.

2. When removing footwear in every 5,000 flexes use caution not to drip water into footwear. If water contaminates footwear, discontinue the test, dry footwear and retest**.**

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(a) Oxford



(b) Boot

Figure 1. Location of testing line and high water mark for **adult** shoes with the lowest point of the topline less than 195mm.

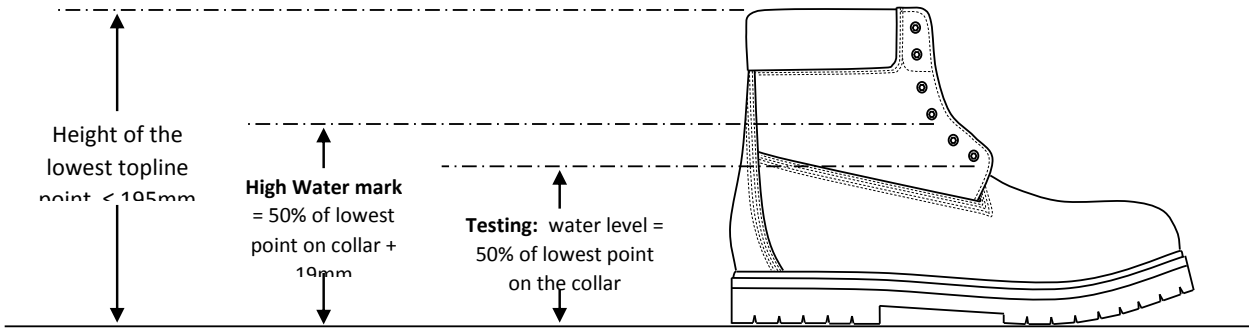


Figure 2. Location of testing line and high water mark for **kids** shoes with the lowest point of the topline less than 195mm.

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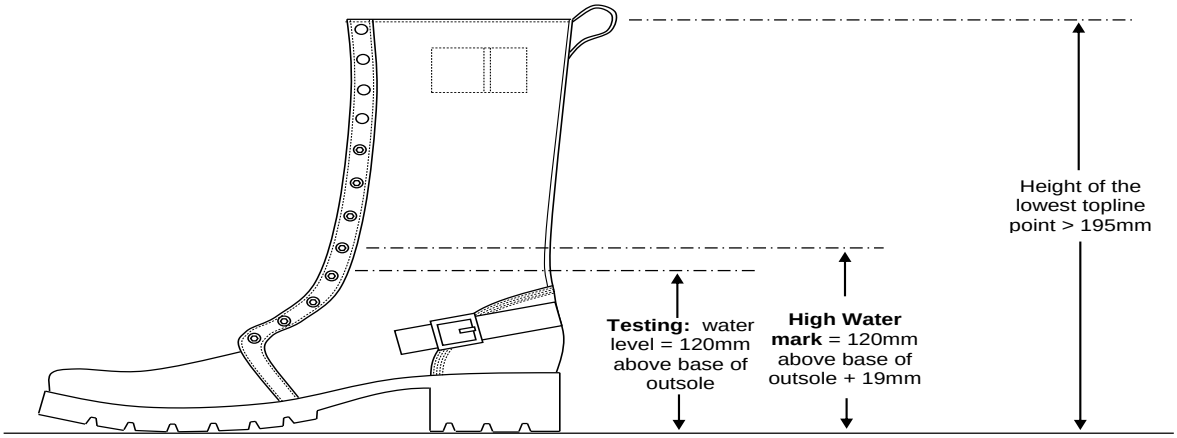


Figure 3. Location of testing line and high water mark for **adult** and **kids** shoes with the lowest point of the topline greater than 195mm.

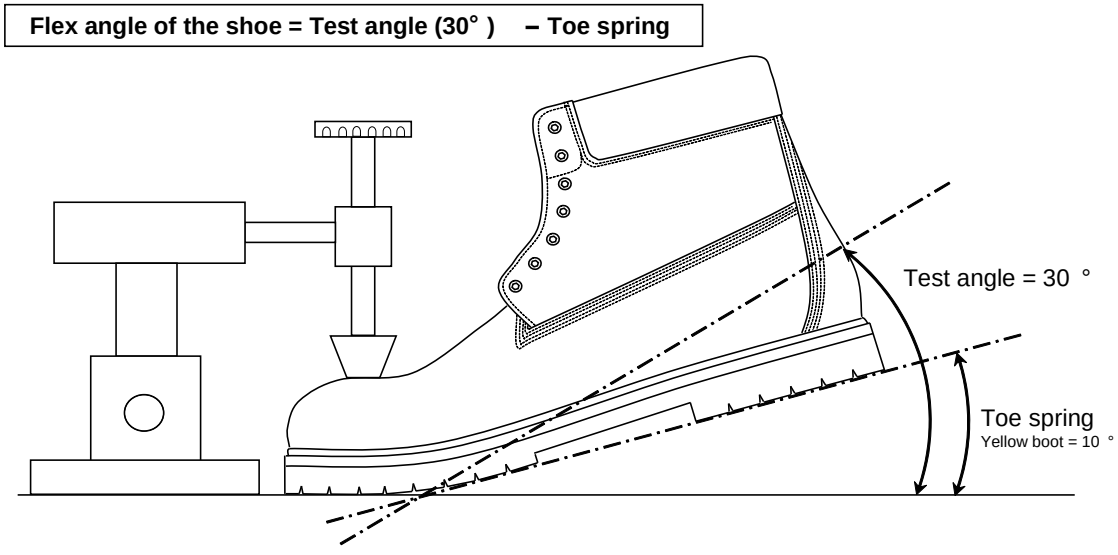
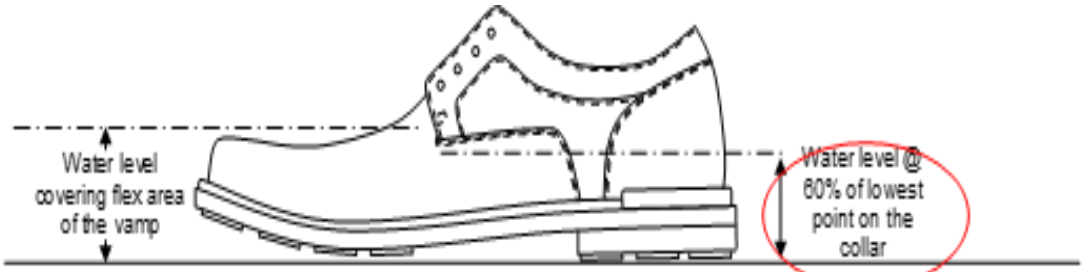


Figure 4. Diagram of testing set up. The test angle should always be set to 30° and the heel should be allowed to float to account for the toe spring.

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(a) Water level for flex testing set at 60% of lowest point on the collar because it is the lower of the two water level measurements.



(b) Water level for flex testing set at just above the flex area of the vamp because it is the lower of the two water level measurements.

Figure 5. Diagram of water levels to be used for flex testing.